

# Yixiu Xiao

*PhD Candidate in Mathematics, SJTU*
*Visiting Research Student, UNSW Sydney*

## Personal details


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## Education

Apr 2025–Oct 2026 **Visiting Research Student (Joint Training Program)**, *The University of New South Wales (UNSW Sydney)*, Sydney, Australia
Supervisor: Prof. Igor E. Shparlinski.

Sep 2021–expected Mar 2027 **PhD Candidate in Mathematics**, *Shanghai Jiao Tong University*, Shanghai, China
Supervisor: Prof. Hongze Li.

Sep 2017–Jun 2021 **BSc in Mathematics and Applied Mathematics**, *Northwest University*, Xi'an, China

## Research Interests

**Analytic Number Theory**

- Exponential sums, character sums, and bilinear forms
- Kloosterman and Salié sums
- Sieve methods, large-sieve inequalities, and the distribution of primes
- Equidistribution modulo one and discrepancy theory
- Diophantine equations, the Hardy–Littlewood circle method, and the delta method

## Current Research Directions

- Squarefree and almost-prime values of norms of $\mathrm{SL}_2(\mathbb{Z})$ matrices
- Sign changes of Kloosterman sums over square-free almost-prime moduli
- Moment estimates and discrepancy for sums of square roots modulo one
- Large-sieve inequalities for quadratic character sums over short intervals
- Shifted bilinear sums of Salié sums and modular square roots of shifted primes
- Least non-trivial zeros of cubic forms and systems of additive cubic equations

## Publications

### Published

**Yixiu Xiao** and Hongze Li. **Least zero of pairs of additive cubic equations**. *Forum Mathematicum* 38 (2026), no. 5, 1569–1598. DOI: 10.1515/forum-2025-0367.

**Yixiu Xiao** and Hongze Li. **Least Zero of a Cubic Form**. *Frontiers of Mathematics* 21 (2026), no. 5, 1043–1069. DOI: 10.1007/s11464-024-0113-6.

### Preprints (arXiv)

**Yixiu Xiao** and Hongze Li. **Selberg sieve weights and sign changes of Kloosterman sums. II. Square-free moduli with at most four prime factors**. *arXiv*: 2609.24255.

**Yixiu Xiao** and Hongze Li. **Selberg sieve weights and sign changes of Kloosterman sums. I. Uniform asymptotics for shifted weights**. *arXiv*: 2609.24248.

Igor E. Shparlinski and **Yixiu Xiao**. **Arithmetic structure of $L_2$-norms of $\mathrm{SL}_2(\mathbb{Z})$ matrices**. *arXiv*: 2609.21460.

**Yixiu Xiao**. **Kloosterman sign changes with moduli having at most five prime factors**. *arXiv*: 2607.10862.

**Yixiu Xiao**. **Moment Estimates and Discrepancy for Sums of Square Roots Modulo One**. *arXiv*: 2606.28986.

Marc Munsch, Igor E. Shparlinski, Yu-Chen Sun, and **Yixiu Xiao**. **Large sieve inequality for sums of Legendre symbols over short intervals**. *arXiv*: 2604.23661.

Igor E. Shparlinski and **Yixiu Xiao**. **Shifted bilinear sums of Salié sums and the distribution of modular square roots of shifted primes**. *arXiv*: 2601.10113.

## Scholarships and Funding

2026 **PhD Support Scheme**, School of Mathematics and Statistics, UNSW Sydney.

2024 **China Scholarship Council (CSC) Scholarship.** Support for 24 months of visiting doctoral research in Australia, April 2025–March 2027.

## Talks

25 Aug 2026 **Moment Estimates and Discrepancy for Sums of Square Roots Modulo One**, FOUVRY–73 Summer School (short talk), Bernoulli Center, EPFL, Lausanne, Switzerland.

5 Aug 2026 **Moment Estimates and Discrepancy for Sums of Square Roots Modulo One**, UNSW Number Theory Seminar, UNSW Sydney.

10 Dec 2025 **Least zero of pairs of additive cubic equations**, AustMS Annual Meeting, La Trobe University, Melbourne, Australia.

9 Sep 2025 **Diophantine equations and cubic forms**, Prime Pals / Number Theory Group Seminar (internal), UNSW Canberra.

## Research Experience

**Research Assistant**, Prof. Igor E. Shparlinski, School of Mathematics and Statistics, UNSW Sydney, 7 April 2026–present.

Conducting research in analytic number theory, including character sums, large-sieve inequalities, Kloosterman and Salié sums, and arithmetic distribution problems.

**Research Assistant**, Prof. Hongze Li, Shanghai Jiao Tong University, 2021–present.

Conducting research on exponential sums, Diophantine equations, cubic forms, and the Hardy–Littlewood circle method, including proof development and manuscript preparation.

## Teaching

Term 2, 2026 **MATH1231 Mathematics 1B**, Course Marker, UNSW Sydney.

Term 1, 2026 **MATH2018 Engineering Mathematics 2D**, Course Marker, UNSW Sydney.

Term 1, 2026 **MATH1141 Higher Mathematics 1A**, Course Marker, UNSW Sydney.

2025–2026 **MATH8000 Student Support**, Examination Invigilation and Marking, UNSW Sydney. Term 3, 2025; Terms 1–2, 2026.

2022–2024 **Elementary Number Theory (Summer Term)**, Teaching Assistant, Shanghai Jiao Tong University.

2021–2023 **Mathematical Analysis**, Teaching Assistant, Shanghai Jiao Tong University, Fall terms.